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Zincation and Magnesiation of Functionalized Silylated Cyanohydrins Using TMP-Bases

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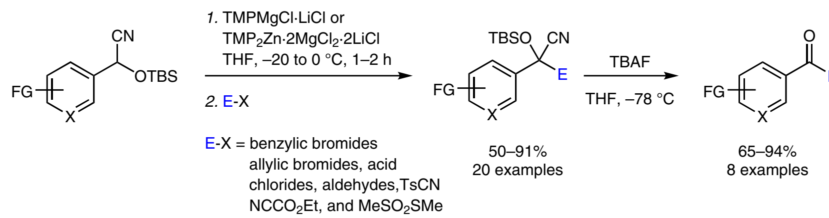

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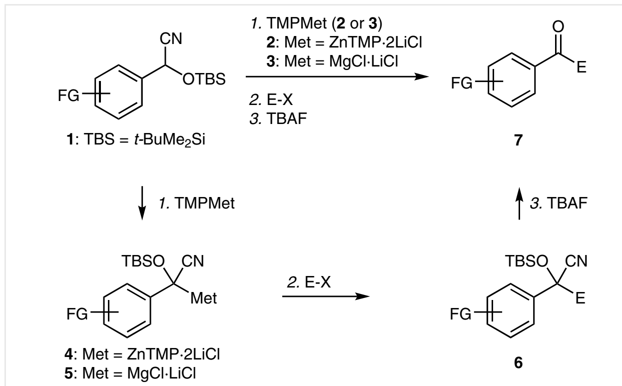
Abstract Polyfunctional silylated cyanohydrins are readily magnesiated or zincated with TMPMgCl·LiCl or $TMP_2Zn·2MgCl_2·2LiCl$ leading to the corresponding metallated derivatives. These Mg- or Zn-derivatives react with various electrophiles such as benzylic bromides, allylic bromides, acid chlorides, aldehydes, $NCCO_2Et$, or $MeSO_2SMe$. Subsequently, TBAF-deprotection provides the corresponding keto or 1,2-diketo derivatives.

Key words metalation, umpolung, organomagnesium reagents, organozinc reagents, benzylation

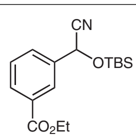
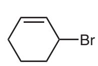
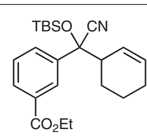
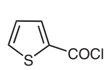
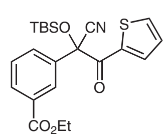
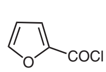
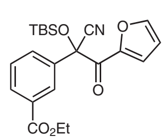
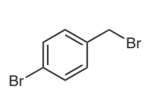
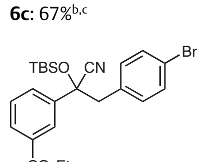
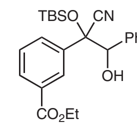
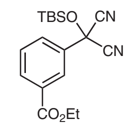
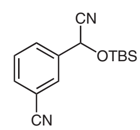
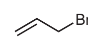
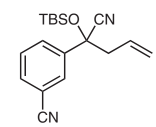
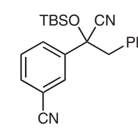
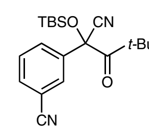
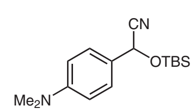
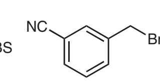
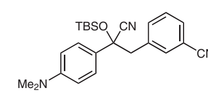
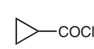
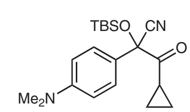
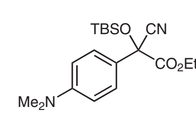
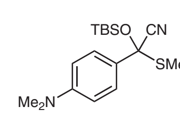
The preparation of complex organic molecules has been rationalized by using retrosynthetic analyses involving synthons of opposite polarity.[1] In this respect, synthons with inverted polarity (synthons with umpolung)[2] are of special importance and considerable studies have been made for the elaboration of synthetic equivalents of acyl anions.[3] Pioneering work of Stork in 1971 demonstrated that protected cyanohydrins can be readily deprotonated by LDA at –78 °C and further functionalized by reactions with various electrophiles.[4] Since highly reactive lithium bases are required for the generation of lithium cyanohydrin derivatives, very few functional groups are tolerated.[5] Recently, we have developed a range of powerful Mg- and Zn-TMP-bases (TMP = 2,2,6,6-tetramethylpiperidyl), which perform selective metalations on functionalized aromatic and heterocyclic substrates under mild conditions.[6] We have envisioned to use these more chemoselective TMP-bases for the preparation of a range of magnesiated and zincated functionalized silylated cyanohydrin derivatives. Herein, we wish to report a highly chemoselective metalation of functionalized silylated cyanohydrins of type **1** using either $TMP_2Zn·2MgCl_2·2LiCl$ (**2**, abbreviated $TMP_2Zn·2LiCl$)[7] or TMPMgCl·LiCl (**3**)[8] leading to the corresponding metalated cyanohydrins **4** (Met = ZnTMP·2LiCl) and **5** (Met = MgCl·LiCl) (Scheme 1). After trapping of **4** or **5** with a range of electrophiles (E–X), the corresponding polyfunctionalized cyanohydrin derivatives **6** were obtained and converted by treatment with tetrabutylammonium fluoride (TBAF)[9] to the corresponding polyfunctional keto derivatives **7**.



Scheme 1 Functionalization of silylated protected cyanohydrins **1** using TMP-bases and subsequent deprotection with TBAF affording the keto derivatives **7**

Thus, the silylated cyanohydrin **1a**, which was readily prepared from the corresponding aldehyde[10] (t-$BuMe_2SiCN$, cat. CsF, MeCN, 25 °C, ca. 50–98% yield), was treated with $TMP_2Zn·2LiCl$ (**2**; 1.1 equiv) in THF at –20 °C. After 2 hours of reaction time, a complete zincation was achieved as indicated by copper-catalyzed allylation of reaction aliquots with allyl bromide. The zincated cyanohydrin derivative of type **4** was allylated with 3-bromocyclohexene in the presence of 20% CuCN·2LiCl[11] leading to the allylated cyanohydrin **6a** in 54% yield (Table 1, entry 1). Acylations of zincated cyanohydrins with acid chlorides were best performed in the presence of 20% CuCN·2LiCl[11] and provided the keto derivatives **6b,c** in 62–67% yield (entries 2, 3). Also, benzylation of **1a** was achieved using 10% CuBr[12] leading to the corresponding product **6d** in 85% yield (entry 4). Quenching of zinc intermediates of type **4** with less reactive electrophiles such as an aldehyde provided low yields. However, the reaction of **1a** with TMPMgCl·LiCl (**3**; 1.3 equiv) in THF at –20 °C for 2 hours afforded a magnesiated cyanohydrin of type **5**. Its reaction with benzaldehyde or TsCN provided the expected cyanohydrin derivatives **6e**,**f** in 50–79% yield (entries 5, 6). Similarly, the cyano-substituted silylated cyanohydrin **1b** was zincated with $TMP_2Zn·2LiCl$ (**2**; 1.1 equiv, –20 °C, 2 h) leading to the corresponding zinc derivative of type **4**. After allylation, benzylation, and acylation, the desired cyanohydrin derivatives **6g**–**i** were obtained in 67–74% yield (entries 7–9). Also, a silylated cyanohydrin bearing a dimethylamino substituent **1c** was readily zincated or magnesiated with **2** (1.1 equiv, –20 °C, 2 h) or **3** (1.3 equiv, –20 °C, 2 h) leading to metalated derivatives of type **4** and **5**. The zincated species derived from **1c** were then quenched with various electrophiles such as 3-cyanobenzyl bromide and cyclopropanecarbonyl chloride in the presence of catalytic amount of copper providing the desired cyanohydrin derivatives **6j**,**k** in 83–85% yield (entries 10, 11). The magnesiated species derived from **1c** reacted with ethyl cyanoformate and $MeSO_2SMe$ providing polyfunctional cyanohydrin derivatives **6l**, **m** in 74–89% yield (entries 12, 13).

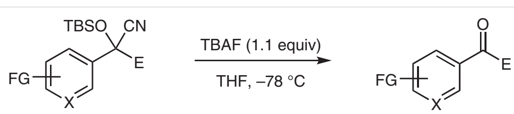
Table 1 Preparation of Functionalized Protected Cyanohydrins of Type **6**

Entry	Substrate **1**	Electrophile (E–X)	Product/Yield[a]
1	 **1a**		 **6a**: 54%[b,c]
2	**1a**		 **6b**: 62%[b,c]
3	**1a**		 **6c**: 67%[b,c]
4	**1a**		 **6d**: 85%[b,d]
5	**1a**	PhCHO	 **6e**: 79%[e]
6	**1a**	TsCN	 **6f**: 50%[e]
7	 **1b**		 **6g**: 67%[b,c]
8	**1b**	BnBr	 **6h**: 74[b,d]
9	**1b**	*t*-BuCOCl	 **6i**: 74%[b,c]
10	 **1c**		 **6j**: 83%[b,d]
11	**1c**		 **6k**: 85%[b,c]
12	**1c**	$NCCO_2Et$	 **6l**: 89%[e]
13	**1c**	$MeSO_2SMe$	 **6m**: 74%[e]

[a] Yield of isolated, analytically pure product.
[b] $TMP_2Zn·2LiCl$ (**2**; 1.1 equiv) was used at –20 °C for 2 h.
[c] CuCN·2LiCl (20 mol%) was used.
[d] CuBr (10 mol%) was used.
[e] TMPMgCl·LiCl (**3**; 1.3 equiv) was used at –20 °C for 2 h.

Subsequently, we turned our attention to heterocyclic silylated cyanohydrins. Thus, the 3-pyridyl cyanohydrin derivative **1d** was smoothly zincated with $TMP_2Zn\cdot 2LiCl$ (**2**; 1.1 equiv) at 0 °C for 2 h. Quenching with ethyl 2-(bromomethyl)acrylate[13] in the presence of 20% $CuCN\cdot 2LiCl$[11] furnished the allylated product **6n** in 60% yield (Table 2, entry 1). Copper-catalyzed benzylation with 3-cyanobenzyl bromide provided the cyanohydrin **6o** in 75% yield (entry 2). 2-Bromopyridine (**1e**) was also metalated using $TMP_2Zn\cdot 2LiCl$ (**2**; 1.1 equiv) at 0 °C for 2 hours and the zincated species was quenched with cyclopropanecarbonyl chloride in the presence of 20% $CuCN\cdot 2LiCl$[11] to give ketone **6p** in 91% yield (entry 3). Silylated cyanohydrin pyridine **1f** was readily metalated using $TMP_2Zn\cdot 2LiCl$ (**2**; 1.1 equiv) at 0 °C for 1 hour. Its copper-catalyzed acylation produced the expected product **6q** in 83% yield (entry 4). In the case of the protected indole **1g**, the zincated species obtained using $TMP_2Zn\cdot 2LiCl$ (**2**; 1.1 equiv, 0 °C, 2 h) was quenched with allyl bromide in the presence of 20% $CuCN\cdot 2LiCl$[11] or with methyl 4-formylbenzoate giving the desired products **6r**,**s** in 63–72% yield (entries 5, 6). Similarly, the protected uracil derivative **1h** was also deprotonated with $TMP_2Zn\cdot 2LiCl$ (**2**; 1.1 equiv, –20 °C, 2 h) and the corresponding metalated species was reacted with 2,6-dichorobenzyl bromide using 10% CuBr[12] as catalyst providing the complex pyrimidine **6t** in 79% yield (entry 7).

In order to show the utility of this methodology, the obtained polyfunctionalized silylated cyanohydrins **6** were then converted into the corresponding polyfunctional keto derivatives **7** using tetrabutylammonium fluoride (TBAF, Scheme 2) in THF.

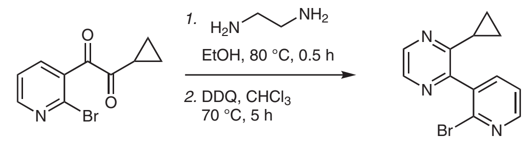


6 7

Scheme 2 TBAF deprotection of functionalized silylated cyanohydrins **6**

Thus, the treatment of silylated protected cyanohydrin derivatives **6d**,**h**,**k**–**o** (Table 3, entries 1–7) with TBAF (1.1 equiv, –78 °C) in THF for 0.5 hour (except for **6p**, entry 8, where a reaction time of 1.5 h was required) produced the corresponding ketones **7a**–**h** in 65–94% yield.

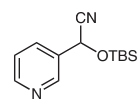
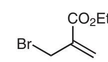
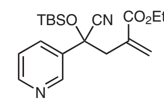
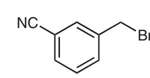
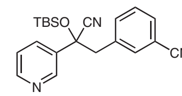
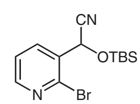
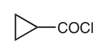
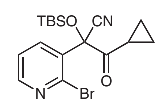
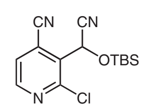
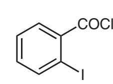
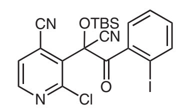
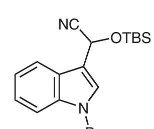
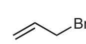
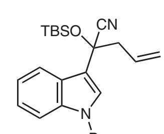
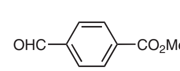
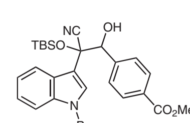
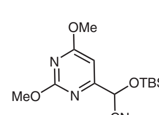
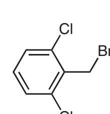
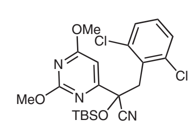
Additionally, the previously prepared 1,2-dione **7h** could be further used in the synthesis of a new pyrazine. Thus, the reaction of **7h** with ethylenediamine (1.5 equiv) in EtOH at 80 °C for 0.5 hour gave an adduct, which was readily oxidized with 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ, 2.0 equiv, 70 °C, 5 h) in chloroform to afford the desired pyrazine **8** in 32% yield over 2 steps (Scheme 3).



7h **8**: 32% over 2 steps

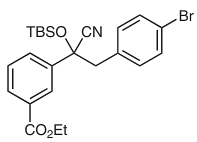
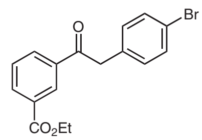
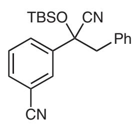
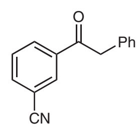
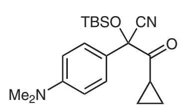
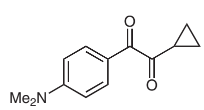
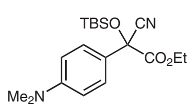
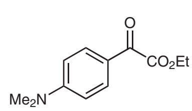
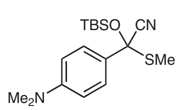
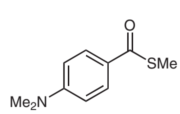
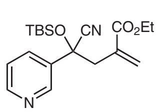
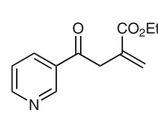
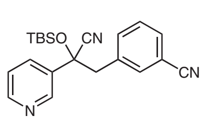
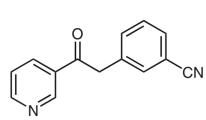
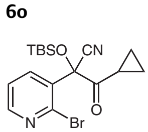
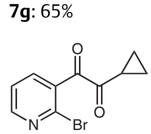
Scheme 3 Synthesis of pyrazine **8** from the keto derivative **7h**

Table 2 Synthesis of Heterocyclic Functionalized Protected Cyanohydrins of Type **6**

Entry	Substrate **1**	Electrophile (E–X)	Product/Yield[a]
1	 **1d**		 **6n**: 60%[b]
2	**1d**		 **6o**: 75%[c]
3	 **1e**		 **6p**: 91%[b]
4	 **1f**		 **6q**: 83%[b]
5	 **1g**		 **6r**: 72%[b]
6	**1g**		 **6s**: 63%
7	 **1h**		 **6t**: 79%[c]

[a] Yield of isolated, analytically pure product.
[b] $CuCN\cdot 2LiCl$ (20 mol%) was used.
[c] CuBr (10 mol%) was used.

Table 3 Synthesis of Poyfunctional Keto Derivatives of Type **7**

Entry	Substrate **6**	Product/Yield[a]
1	 **6d**	 **7a**: 71%
2	 **6h**	 **7b**: 81%
3	 **6k**	 **7c**: 81%
4	 **6l**	 **7d**: 86%
5	 **6m**	 **7e**: 94%
6	 **6n**	 **7f**: 83%
7	 **6o**	 **7g**: 65%
8	 **6p**	 **7h**: 90%

[a] Yield of isolated, analytically pure product.

In summary, we have shown that $TMP_2Zn{\cdot}2LiCl$ (**2**) and TMPMgCl·LiCl (**3**) are excellent bases in the metalation of protected silylated cyanohydrins. These metalated silylated cyanohydrins are masked acyl anion equivalents bearing sensitive functionalities and react readily with a range of useful electrophiles. Subsequent TBAF deprotection provides polyfunctional keto derivatives.

Unless otherwise indicated, all reactions were carried out with magnetic stirring and in flame-dried glassware under argon. Syringes used to transfer reagents and solvents were purged with argon prior to use. Reactions were monitored by GC or GC-MS or TLC. TLC was performed using aluminum plates covered with silica gel (Merck 60, F-254) and visualized by UV detection. Purification by column chromatography was performed using Merck silica gel 60 (40–63 mm 230–400 mesh ASTM from Merck) or using preparative high-performance liquid chromatography (HPLC). THF was continuously refluxed and freshly distilled from Na benzophenone ketyl under N_2. Yields refer to isolated yields of compounds estimated to be >95% pure as determined by ^1H NMR spectroscopy (25 °C) and capillary GC analysis. Melting points were measured using a Büchi B 540 apparatus and are uncorrected. NMR spectra were recorded in $CDCl_3$. Mass spectra and HRMS were recorded on Finnigan MAT 90 or Finnigan MAT 95 instruments using electron ionization (EI). IR spectra were recorded on a PerkinElmer spectrum BX-59343 instrument. For detection, a Smiths Detection DuraSamplIR II Diamond ATR sensor was used. Gas chromatographic analyses were performed on a Hewlett-Packard 6890 instrument (5% phenylmethylpolysiloxane; length: 15 m, diameter: 0.25 mm; film thickness: 0.25 μm). 2,2,6,6-Tetramethylpiperidine (TMPH) was distilled prior to use, and CuCN, $ZnCl_2$, and LiCl were obtained from Merck.

CuCN·2LiCl Solution[11]

CuCN·2LiCl solution (1.0 M in THF) was prepared by drying CuCN (7.17 g, 80 mmol) and LiCl (6.77 g, 160 mmol) in a Schlenk flask under vacuum at 140 °C for 5 h. After cooling, anhyd THF (80 mL) was added and stirring was continued until all salts were dissolved (24 h).

$ZnCl_2$ Solution

$ZnCl_2$ solution (1.0 M in THF) was prepared by drying $ZnCl_2$ (136.3 g, 100 mmol) in a Schlenk flask under vacuum at 140 °C for 5 h. After cooling, anhyd THF (100 mL) was added and stirring was continued until all salts were dissolved (12 h).

i-PrMgCl·LiCl[14]

Mg turnings (110 mmol) and anhyd LiCl (100 mmol) were placed in an argon-flushed flask, and THF (50 mL) was added. A solution of *i*-PrCl (100 mmol) in THF (50 mL) was slowly added at 25 °C. The reaction started within a few minutes. After the addition, the reaction mixture was stirred for 12 h at 25 °C. The gray solution of *i*-PrMgCl·LiCl was cannulated into another argon-filled flask and removed in this way from excess Mg. *i*-PrMgCl·LiCl was titrated with I_2 prior to use.

TMPMgCl·LiCl (3)[8a]

In a dry and argon-flushed Schlenk flask, TMPH (14.8 g, 105 mmol) was added to *i*-PrMgCl·LiCl (71.4 mL, 100 mmol, 1.40 M in THF) at 25 °C and the mixture was stirred for 3 days at 25 °C. The freshly prepared TMPMgCl·LiCl (**3**) was titrated prior to use at 0 °C with benzoic acid using 4-(phenylazo)diphenylamine as indicator.

$TMP_2Zn·2MgCl_2·2LiCl$ (2)[7a]

A flame-dried and argon-flushed Schlenk flask, equipped with a magnetic stirring bar and rubber septum, was charged with TMPMgCl·LiCl (**3**; 348 mL, 400 mmol) and cooled to 0 °C. Then, $ZnCl_2$ (200 mL, 200 mmol, 1.0 M in THF) was added over a period of 15 min. After stirring this mixture for 2 h at 0 °C, the solution of $TMP_2Zn·2MgCl_2·2LiCl$ was concentrated in vacuo. Freshly distilled THF was then slowly added and the solution was stirred until all salts were completely dissolved. The freshly prepared $TMP_2Zn·2MgCl_2·2LiCl$ (**2**) was titrated prior to use at 0 °C with benzoic acid using 4-(phenylazo)diphenylamine as indicator.

Silylated Protected Cyanohydrins 1; General Procedure 1 (GP1)

According to the literature,[10h] an argon flushed round-bottom flask with a magnetic stirring bar and a septum was charged with the aryl carboxaldehyde (1.0 equiv), CsF (0.2 equiv), *t*-$BuMe_2SiCN$ (1.5 equiv) and anhyd MeCN (1.0 M). The reaction mixture was stirred at 25 °C and monitored by GC analysis ($C_{11}H_{24}$ was used as an internal standard).

Zincation of Protected Cyanohydrins 1 with $TMP_2Zn·2MgCl_2·2LiCl$ (2); General Procedure 2 (GP2)

A dry and argon flushed Schlenk flask was charged with a solution of the protected cyanohydrin **1** (1.0 equiv) in anhyd THF (0.25 M). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 1.1 equiv) was added dropwise at the indicated temperature and the reaction mixture was stirred for 1–2 h. The completion of the reaction was checked by TLC analysis of reaction aliquots quenched with allyl bromide in the presence of CuCN·2LiCl in anhyd THF.

Magnesiation of Protected Cyanohydrins 1 with TMPMgCl·LiCl (3); General Procedure 3 (GP3)

A dry and argon-flushed Schlenk flask was charged with a solution of the protected cyanohydrin **1** (1.0 equiv) in anhyd THF (0.25 M). TMPMgCl·LiCl (**3**; 1.3 equiv) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. The completion of the reaction was checked by TLC analysis of reaction aliquots quenched with allyl bromide in the presence of CuCN·2LiCl in anhyd THF.

Deprotection of Silylated Cyanohydrins 6 with TBAF; General Procedure 4 (GP4)

An argon-flushed round-bottom flask with a magnetic stirring bar and a septum was charged with silylated cyanohydrin derivative **6** (1.0 equiv) in anhyd THF (0.07 M) at –78 °C. TBAF (1.1 equiv, 1.0 M in THF) was added dropwise to the solution. The reaction mixture was then stirred at –78 °C. The completion of the reaction was checked by TLC analysis of reaction aliquots.

Ethyl 3-Formylbenzoate

According to the literature,[15] 3-formylbenzoic acid (751 mg, 5.00 mmol) was dissolved in DMF (50 mL), and K_2CO_3 (1.4 g, 10.0 mmol) and EtI (1.6 g, 10.0 mmol) were added to this solution. The reaction mixture was stirred at 60 °C under N_2 atmosphere for 14 h. The resulting solution was quenched with sat. aq NH_4Cl (20 mL), extracted with EtOAc (3 × 50 mL) and dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 96:4 to 9:1) yielding ethyl 3-formylbenzoate as a white solid (561 mg, 63%).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 10.11 (s, 1 H), 8.59 (s, 1 H), 8.29 (d, *J* = 7.6 Hz, 1 H), 8.11 (d, *J* = 7.6 Hz, 1 H), 7.65 (t, *J* = 7.6 Hz, 1 H), 4.46 (q, *J* = 7.0 Hz, 2 H), 1.44 (t, *J* = 7.0 Hz, 3 H).

Ethyl 3-{[(*tert*-Butyldimethylsilyl)oxy](cyano)methyl}benzoate (1a)

According to GP1, ethyl 3-formylbenzoate (866 mg, 4.90 mmol), CsF (148 mg, 0.97 mmol), and *t*-$BuMe_2SiCN$ (1.3 g, 7.30 mmol) were dissolved in anhyd MeCN (4.9 mL). The reaction mixture was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (20 mL) and extracted with EtOAc (3 × 50 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5 to 9:1) yielding **1a** as a colorless oil (1.4 g, 90%).

IR (Diamond-ATR, neat): 2932, 2860, 2256, 1719, 1472, 1367, 1255, 1188, 1107, 914, 837, 780 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.15 (br s, 1 H), 8.08 (d, *J* = 7.6 Hz, 1 H), 7.69 (d, *J* = 7.6 Hz, 1 H), 7.52 (t, *J* = 7.6 Hz, 1 H), 5.58 (s, 1 H), 4.41 (q, *J* = 7.0 Hz, 2 H), 1.42 (t, *J* = 7.0 Hz, 3 H), 0.96 (s, 9 H), 0.26 (s, 3 H), 0.18 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 165.8, 136.9, 131.3, 130.4, 130.2, 129.1, 127.2, 118.9, 63.5, 61.3, 25.5, 18.2, 14.3, –5.1, –5.2.

MS (70 eV, EI): *m*/*z* (%) = 304 (5), 262 (100), 219 (64), 207 (26), 202 (46), 191 (8), 163 (12), 133 (13).

HRMS (EI): *m*/*z* [M – Me]$^+$ calcd for $C_{16}H_{22}NO_3Si$: 304.1369; found: 304.1364.

3-{[(*tert*-Butyldimethylsilyl)oxy](cyano)methyl}benzonitrile (1b)

According GP1, 3-cyanobenzaldehyde (525 mg, 4.00 mmol), CsF (122 mg, 0.80 mmol), and *t*-$BuMe_2SiCN$ (848 mg, 6.00 mmol) were dissolved in anhyd MeCN (4.0 mL). The reaction mixture was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (20 mL) and extracted with EtOAc (3 × 50 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **1b** as a white solid (1.1 g, 98%); mp 40.6–42.0 °C.

IR (Diamond-ATR, neat): 2932, 2860, 2234, 1471, 1363, 1256, 1149, 1086, 902, 835, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.78 (s, 1 H), 7.70 (d, *J* = 8.2 Hz, 1 H), 7.73 (d, *J* = 8.2 Hz, 1 H), 7.57 (t, *J* = 8.2 Hz, 1 H), 5.56 (s, 1 H), 0.96 (s, 9 H), 0.27 (s, 3 H), 0.20 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 138.1, 132.8, 130.2, 129.8, 129.5, 118.3, 118.0, 113.2, 63.0, 25.4, 18.1, –5.2, –5.3.

MS (70 eV, EI): *m*/*z* (%) = 272 (0.5), 246 (10), 215 (100), 147 (9), 84 (33), 73 (21), 57 (13).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{15}H_{20}N_2OSi$: 272.1345; found: 272.1324.

2-[(*tert*-Butyldimethylsilyl)oxy]-2-[4-(dimethylamino)phenyl]-acetonitrile (1c)

According to GP1, 4-(dimethylamino)benzaldehyde (447 mg, 3.00 mmol), CsF (91 mg, 0.60 mmol), and *t*-$BuMe_2SiCN$ (636 mg, 4.50 mmol) were dissolved in anhyd MeCN (3.0 mL). The reaction was mixture stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (20 mL) and extracted with EtOAc (3 × 50 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 96:4) yielding **1c** as a yellow oil (766 mg, 88%).

IR (Diamond-ATR, neat): 2930, 2858, 2255, 1613, 1524, 1471, 1360, 1254, 1166, 1064, 904, 835, 778 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.32 (d, *J* = 8.6 Hz, 2 H), 6.73 (d, *J* = 8.6 Hz, 2 H), 5.43 (s, 1 H), 2.99 (s, 6 H), 0.93 (s, 9 H), 0.20 (s, 3 H), 0.12 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 150.9, 127.5, 119.7, 112.2, 64.0, 40.4, 25.6, 18.1, –5.1.

MS (70 eV, EI): *m*/*z* (%) = 290 (11), 160 (15), 159 (100), 148 (12), 75 (7).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{16}H_{26}N_2OSi$: 290.1814; found: 290.1818.

2-[(*tert*-Butyldimethylsilyl)oxy]-2-(pyridin-3-yl)acetonitrile (1d)

According to GP1, 3-pyridinecarboxaldehyde (428 mg, 4.00 mmol), CsF (122 mg, 0.80 mmol), and *t*-$BuMe_2SiCN$ (848 mg, 6.00 mmol) were dissolved in anhyd MeCN (4.0 mL). The reaction mixture was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (20 mL) and extracted with EtOAc (3 × 50 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1 to 8:2 + 2% Et_3N) yielding **1d** as a colorless oil (903 mg, 91%).

IR (Diamond-ATR, neat): 2931, 2859, 1592, 1579, 1427, 1363, 1255, 1089, 920, 835, 780 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.69 (s, 1 H), 8.63 (d, *J* = 4.5 Hz, 1 H), 7.81 (d, *J* = 7.8 Hz, 1 H), 7.35 (dd, *J* = 7.8, 4.5 Hz, 1 H), 5.56 (s, 1 H), 0.92 (s, 9 H), 0.24 (s, 3 H), 0.16 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 150.5, 147.6, 133.7, 132.3, 123.6, 118.3, 62.0, 25.4, 18.0, –5.2, –5.3.

MS (70 eV, EI): *m*/*z* (%) = 248 (2), 191 (100), 117 (7), 84 (24), 75 (9).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{13}H_{20}N_2OSi$: 248.1345; found: 248.1345.

2-Bromo-3-pyridinecarboxaldehyde

According to the literature,[16] a dry and argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with a solution of LDA (30.0 mmol, 0.5 M in THF) and cooled to –78 °C. 2-Bromopyridine (7.9 g, 10.0 mmol) was added dropwise to the cooled solution. The resulting mixture was stirred for 1 h at –78 °C. DMF (2.9 g, 40.0 mmol) was then added and the mixture stirred for 1 h at –78 °C. The resulting solution was quenched with sat. aq NH_4Cl (40 mL), extracted with EtOAc (3 × 80 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 8:2 + Et_3N 2%) yielding 2-bromo-3-pyridine-carboxaldehyde as a colorless oil (1.0 g, 54%).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 10.34 (1 H, s), 8.60 (dd, *J* = 4.5, 2.1 Hz, 1 H), 8.19 (dd, *J* = 7.9, 2.1 Hz, 1 H), 7.46 (dd, *J* = 7.9, 4.5 Hz, 1 H).

2-(2-Bromopyridin-3-yl)-2-[(*tert*-butyldimethylsilyl)oxy]acetonitrile (1e)

According to GP1, 2-bromo-3-pyridinecarboxaldehyde (430 mg, 2.30 mmol), CsF (70 mg, 0.46 mmol), and *t*-$BuMe_2SiCN$ (49 mg, 3.5 mmol) were dissolved in anhyd MeCN (2.3 mL). The reaction mixture was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (10 mL) and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **1e** as a colorless oil (670 mg, 86%).

IR (Diamond-ATR, neat): 2931, 2859, 1563, 1401, 1363, 1256, 1103, 1049, 938, 835, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.43 (dd, *J* = 4.8, 1.6 Hz, 1 H), 8.02 (dd, *J* = 7.7, 1.6 Hz, 1 H), 7.41 (dd, *J* = 7.7, 4.8 Hz, 1 H), 5.69 (s, 1 H), 0.95 (s, 9 H), 0.30 (s, 3 H), 0.20 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 150.7, 140.9, 136.7, 133.4, 123.5, 117.5, 62.8, 25.5, 18.1, –5.2, –5.3.

MS (70 eV, EI): *m*/*z* (%) = 312 (1), 271 (100), 270 (15), 190 (51), 139 (29), 137 (31), 84 (13).

HRMS (EI): *m*/*z* [M – Me]$^+$ calcd for $C_{13}H_{19}BrN_2OSi$: 312.0294; found: 312.0253.

2-Chloro-4-iodonicotinaldehyde

According to the literature,[17] a dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with a solution of LDA (18.0 mmol, 1.0 M in THF) and cooled to –78 °C. 2-Chloro-3-iodopyridine (3.1 g, 13.0 mmol) was added dropwise to the cooled solution. The resulting mixture was stirred for 3 h at –78 °C. Ethyl formate (2.5 g, 34.0 mmol) was then added and stirred for 1.5 h at –78 °C. The resulting solution was quenched with sat. aq NH_4Cl (80 mL), extracted with EtOAc (3 × 120 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1 to 8:2 + Et_3N 2%) yielding 2-chloro-4-iodonicotinaldehyde as a yellow solid (1.8 g, 52%).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 10.19 (s, 1 H), 8.07 (d, *J* = 5.1 Hz, 1 H), 7.94 (d, *J* = 5.1 Hz, 1 H).

2-(*tert*-Butyldimethylsilyloxy)-2-(2-chloro-4-iodopyridin-3-yl)acetonitrile

According to GP1, 2-chloro-4-iodonicotinaldehyde (989 mg, 3.70 mmol), CsF (112 mg, 0.74 mmol), and *t*-$BuMe_2SiCN$ (784 mg, 5.60 mmol) were dissolved in anhyd MeCN (3.7 mL). The reaction mixture was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (20 mL) and extracted with EtOAc (3 × 40 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1 + Et_3N 2%) yielding 2-(*tert*-butyldimethylsilyloxy)-2-(2-chloro-4-iodopyridin-3-yl)acetonitrile as a light yellow solid (1.5 g, 98%); mp 83.4–85.2 °C.

IR (Diamond-ATR, neat): 2955, 2859, 1636, 1548, 1472, 1337, 1302, 1255, 1186, 1066, 939, 833, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.99 (d, *J* = 5.2 Hz, 1 H), 7.85 (d, *J* = 5.2 Hz, 1 H), 6.24 (s, 1 H), 0.94 (s, 9 H), 0.31 (s, 3 H), 0.13 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 150.3, 149.9, 135.5, 132.5, 116.7, 110.3, 65.2, 25.4, 18.0, –4.9, –5.1.

MS (70 eV, EI): *m*/*z* (%) = 353 (28), 351 (79), 323 (52), 296 (4), 226 (35), 224 (100), 209 (12), 150 (6).

HRMS (EI): *m*/*z* [M – C_4H_9]$^+$ calcd for $C_9H_9ClIN_2OSi$: 350.9217; found: 350.9210.

3-[(*tert*-Butyldimethylsilyloxy)(cyano)methyl]-2-chloroisonicotinonitrile (1f)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with 2-(*tert*-butyldimethylsilyloxy)-2-(2-chloro-4-iodopyridin-3-yl)acetonitrile (2.3 g, 5.70 mmol) in THF (23 mL), and *i*-PrMgCl·LiCl (5.0 mL, 5.90 mmol, 1.2 M) was added at –60 °C. After 0.5 h, GC analysis of hydrolyzed reaction aliquot showed full consumption of the starting material. A solution of tosyl cyanide (1.6 g, 8.60 mmol) in THF (5 mL) was then added dropwise at –78 °C and the mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (30 mL), extracted with EtOAc (3 × 60 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1 + Et_3N 2%) yielding **1f** as a colorless oil (1.2 g, 67%).

IR (Diamond-ATR, neat): 2932, 2860, 1633, 1575, 1472, 1364, 1257, 1166, 1102, 1072, 940, 839 cm^{-1}.

^1H NMR (400 MHz, $CDCl_3$): δ = 8.63 (d, *J* = 4.9 Hz, 1 H), 7.65 (d, *J* = 4.9 Hz, 1 H), 6.04 (s, 1 H), 0.96 (s, 9 H), 0.34 (s, 3 H), 0.21 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ =151.0, 150.7, 132.1, 126.9, 122.7, 116.0, 113.6, 60.4, 25.4, 18.2, -5.3.

MS (70 eV, EI): *m*/*z* (%) = 306 (1), 292 (7), 250 (100), 223 (12), 208 (19), 93 (20).

HRMS (EI): *m*/*z* $[M – H]^+$ calcd for $C_{14}H_{17}ClN_3OSi$: 306.0829; found: 306.0821.

***tert*-Butyl 3-Formyl-1*H*-indole-1-carboxylate**

According to the literature,[18] di-*tert*-butyl dicarbonate (2.6 g, 12.0 mmol) and Et_3N (1.8 mL, 13.0 mmol), followed by DMAP (122 mg, 1.00 mmol) were added to a solution of indole-3-carboxaldehyde (1.5 g, 10.0 mmol) in anhyd CH_2Cl_2 (50 mL). The reaction mixture was then stirred at 25 °C for 12 h. The resulting mixture was diluted with CH_2Cl_2 and washed with sat. aq NH_4Cl (50 mL). The product was extracted with CH_2Cl_2(2 × 80 mL) and dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding *tert*-butyl 3-formyl-1*H*-indole-1-carboxylate as a white solid (2.3 g, 96%).

^1H NMR (400 MHz, $CDCl_3$): δ = 10.11 (s, 1H), 8.29-8.32 (m, 1H), 8.23 (s, 1H), 8.14 (d, J = 7.5 Hz, 1H), 7.45-7.30 (m, 2H), 1.71 (s, 9H).

***tert*-Butyl 3-[(*tert*-Butyldimethylsilyloxy)(cyano)methyl]-1*H*-indole-1-carboxylate (1g)**

According to GP1, *tert*-butyl 3-formyl-1*H*-indole-1-carboxylate (2.4 g, 9.60 mmol), CsF (288 mg, 1.90 mmol), and *t*-$BuMe_2SiCN$ (2.0 g, 14.4 mmol) were dissolved in anhyd MeCN (9.6 mL). The reaction was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (40 mL) and extracted with EtOAc (3 × 80 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 98:2 to 95:5) yielding **1g** as a colorless oil (3.6 g, 97%).

IR (Diamond-ATR, neat): 2956, 2859, 1739, 1635, 1472, 1369, 1255, 1155, 1089, 909, 838, 767 cm^{-1}.

^1H NMR (400 MHz, $CDCl_3$): δ = 8.18 (d, *J* = 6.9 Hz, 1 H), 7.70–7.74 (m, 2 H), 7.37–7.41 (m, 1 H), 7.31 (t, *J* = 7.4 Hz, 1 H), 5.75 (s, 1 H), 1.70 (s, 9 H), 0.95 (s, 9 H), 0.26 (s, 3 H), 0.19 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 149.3, 135.8, 127.1, 125.2, 124.2, 123.1, 119.4, 118.5, 116.7, 115.5, 84.4, 57.9, 28.1, 25.5, 18.1, –5.1, –5.2.

MS (70 eV, EI): *m*/*z* (%) = 386 (11), 273 (56), 259 (17), 229 (100), 202 (29), 154 (23), 75 (55).

HRMS (EI): *m*/*z* $[M^+]$ calcd for $C_{21}H_{30}N_2O_3Si$: 386.2026; found: 386.2024.

2-(*tert*-Butyldimethylsilyloxy)-2-(2,6-dimethoxypyrimidin-4-yl)acetonitrile (1h)

According to the literature,[19] 2,4-dimethoxypyrimidine (71 mg, 0.50 mmol) was dissolved in anhyd THF (1 mL). $TMPMgCl·LiCl$ (**3**; 0.46 mL, 0.55 mmol) was added dropwise at 25 °C and the reaction mixture was stirred for 0.5 h. DMF (55 mg, 0.75 mmol) was then added at –78 °C and the mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The resulting oil containing the product, 2,6-dimethoxypyrimidine-4-carbaldehyde, and ca. 50% of the 2,4-dimethoxypyrimidine, was dried on the high vacuum line and used in the next step without further purification.

According GP1, the freshly synthesized 2,6-dimethoxypyrimidine-4-carbaldehyde, CsF (7.6 mg, 0.05 mmol), and *t*-$BuMe_2SiCN$ (53 mg, 0.38 mmol) were dissolved in anhyd MeCN (0.5 mL). The reaction mixture was stirred at 25 °C for 12 h. The resulting mixture was diluted with H_2O (5 mL) and extracted with EtOAc (3 × 20 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **1h** as a white solid (37 mg, 26% over 2 steps); mp 90.0–90.9 °C.

IR (Diamond-ATR, neat): 2955, 2860, 1633, 1573, 1463, 1356, 1255, 1129, 1079, 942, 837, 783 cm^{-1}.

^1H NMR (400 MHz, $CDCl_3$): δ = 6.61 (s, 1 H), 5.35 (s, 1 H), 3.98 (s, 3 H), 3.97 (s, 3 H), 0.95 (s, 9 H), 0.25 (s, 3 H), 0.19 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 172.6, 165.7, 165.2, 117.6, 98.0, 64.2, 55.0, 54.1, 25.4, 18.1, –5.3, –5.5.

MS (70 eV, EI): *m*/*z* (%) = 308 (1), 294 (3), 252 (100), 224 (1), 84 (2), 72 (2).

HRMS (EI): *m*/*z* $[M – H]^+$ calcd for $C_{14}H_{22}N_3O_3Si$: 308.1430; found: 308.1408.

Ethyl 3-[(*tert*-Butyldimethylsilyloxy)(cyano)(cyclohex-2-enyl)methyl]benzoate (6a)

According to GP2, ethyl 3-{[(*tert*-butyldimethylsilyl)oxy](cyano)methyl}benzoate (**1a**; 160 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 1.62 mL, 0.55 mmol, 0.34 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.10 mL, 0.10 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. 3-Bromocyclohexene (72 mg, 0.45 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by HPLC yielding **6a** as a colorless oil (97 mg, 54%).

IR (Diamond-ATR, neat): 2932, 2860, 1718, 1636, 1532, 1472, 1368, 1281, 1096, 908, 839, 728 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.21 (s, 1 H), 8.06 (d, *J* = 7.7 Hz, 1 H), 7.73 (d, *J* = 7.7 Hz, 1 H), 7.48 (t, *J* = 7.7 Hz, 1 H), 5.90–6.01 (m, 2 H), 4.33–4.48 (m, 2 H), 2.69 (br s, 1 H), 1.94–2.03 (m, 2 H), 1.65–1.74 (m, 1 H), 1.35–1.46 (m, 4 H), 1.16–1.30 (m, 2 H), 0.95 (s, 9 H), 0.20 (s, 3 H), –0.19 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 165.9, 140.3, 131.4, 130.9, 130.2, 129.9, 128.4, 126.8, 124.9, 119.4, 79.1, 61.2, 49.1, 25.7, 24.9, 24.6, 21.4, 18.3, 14.3, –3.8, –4.2.

MS (70 eV, EI): *m*/*z* (%) = 399 (0.2), 356 (3), 318 (21), 315 (17), 296 (6), 177 (100), 149 (6).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{23}H_{33}NO_3Si$: 399.2230; found: 399.2231.

Ethyl 3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyano-2-oxo-2-(thiophen-2-yl)ethyl]benzoate (6b)

According to GP2, compound **1a** (160 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 1.62 mL, 0.55 mmol, 0.34 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.10 mL, 0.10 mmol, 1.0 M in THF) was added at –40 °C and the mixture stirred for 5 min. 2-Thiophenecarbonyl chloride (95 mg, 0.65 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL) extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5 to 9:1) yielding **6b** as a light yellow oil (133 mg, 62%).

IR (Diamond-ATR, neat): 2932, 2860, 2244, 1720, 1668, 1514, 1409, 1353, 1253, 1186, 1039, 838, 784 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.42 (s, 1 H), 8.08 (d, *J* = 7.7 Hz, 1 H), 7.81 (d, *J* = 7.7 Hz, 1 H), 7.77 (d, *J* = 4.0 Hz, 1 H), 7.67 (d, *J* = 4.0 Hz, 1 H), 7.51 (t, *J* = 7.7 Hz, 1 H), 7.04 (t, *J* = 4.0 Hz, 1 H), 4.39 (qd, *J* = 7.1, 1.9 Hz, 2 H), 1.40 (t, *J* = 7.1 Hz, 3 H), 1.00 (s, 9 H), 0.32 (s, 3 H), 0.20 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 184.4, 165.5, 137.4, 136.3, 136.2, 136.0, 131.5, 130.8, 129.7, 129.3, 128.0, 126.3, 117.9, 80.6, 61.3, 25.8, 18.6, 14.2, –3.8, –3.9.

MS (70 eV, EI): *m*/*z* (%) = 429 (0.3), 414 (8), 372 (70), 223 (4), 149 (8), 177 (72), 111 (100).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{22}H_{27}NO_4SSi$: 429.1430; found: 429.1424.

Ethyl 3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyano-2-(furan-2-yl)-2-oxoethyl]benzoate (6c)

According to GP2, compound **1a** (160 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 1.62 mL, 0.55 mmol, 0.34 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.10 mL, 0.10 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. 2-Furoyl chloride (85 mg, 0.65 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5 to 9:1) yielding **6c** as a light yellow oil (139 mg, 67%).

IR (Diamond-ATR, neat): 2932, 2860, 2240, 1720, 1682, 1561, 1460, 1391, 1266, 1188, 1021, 839, 735 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.38 (br s, 1 H), 8.07 (d, *J* = 7.9 Hz, 1 H), 7.81 (d, *J* = 7.9 Hz, 1 H), 7.59 (s, 1 H), 7.50 (t, *J* = 7.9 Hz, 1 H), 7.28 (d, *J* = 3.3 Hz, 1 H), 6.49 (dd, *J* = 3.3, 1.4 Hz, 1 H), 4.35–4.42 (m, 2 H), 1.40 (t, *J* = 7.1 Hz, 3 H), 0.99 (s, 9 H), 0.25 (s, 3 H), 0.22 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 179.3, 165.5, 147.9, 147.6, 137.0, 131.4, 130.7, 129.8, 129.1, 126.5, 122.5, 117.6, 112.4, 79.4, 61.3, 25.6, 18.5, 14.2, –3.9, –4.0.

MS (70 eV, EI): *m*/*z* (%) = 413 (0.3), 356 (68), 318 (30), 177 (100), 149 (14), 95 (46), 73 (96).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{22}H_{27}NO_5Si$: 413.1658; found: 413.1655.

Ethyl 3-[2-(4-Bromophenyl)-1-(*tert*-butyldimethylsilyloxy)-1-cyanoethyl]benzoate (6d)

According to GP2, compound **1a** (160 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 1.62 mL, 0.55 mmol, 0.34 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuBr[12] (7 mg, 0.05 mmol) and solid 4-bromobenzyl bromide (150 mg, 0.60 mmol) were added at –78 °C. The reaction mixture was stirred at –78 °C for 1 h and at 50 °C for 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5) yielding **6d** as a colorless oil (207 mg, 85%).

IR (Diamond-ATR, neat): 2931, 2859, 2256, 1720, 1591, 1489, 1367, 1275, 1193, 1084, 935, 826, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.19 (s, 1 H), 8.06 (d, *J* = 7.8 Hz, 1 H), 7.63 (d, *J* = 7.8 Hz, 1 H), 7.46 (t, *J* = 7.8 Hz, 1 H), 7.40 (d, *J* = 7.6 Hz, 2 H), 7.02 (d, *J* = 7.6 Hz, 2 H), 4.33–4.48 (m, 2 H), 3.21 (d, *J* = 13.5 Hz, 1 H), 3.11 (d, *J* = 13.5 Hz, 1 H), 1.41 (t, *J* = 7.1 Hz, 3 H), 0.94 (s, 9 H), 0.02 (s, 3 H), –0.09 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 165.8, 140.9, 132.6, 131.1, 131.0, 130.0, 129.6, 128.6, 126.3, 121.8, 119.8, 75.5, 61.2, 51.6, 25.7, 18.3, 14.2, –3.8, –4.3.

MS (70 eV, EI): *m*/*z* (%) = 487 (0.2), 444 (5), 432 (24), 324 (27), 318 (100), 177 (84), 149 (11).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{24}H_{30}BrNO_3Si$: 487.1178; found: 487.1166.

Ethyl 3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyano-2-hydroxy-2-phenylethyl]benzoate (6e)

According to GP3, compound **1a** (160 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). TMPMgCl·LiCl (**3**; 0.54 mL, 0.65 mmol, 1.20 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. Benzaldehyde (64 mg, 0.60 mmol) was added at –78 °C and the reaction mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5) yielding **6e** as a colorless oil (124 mg, 79%).

IR (Diamond-ATR, neat): 2996, 2859, 2255, 1716, 1683, 1472, 1302, 1223, 1116, 1072, 907, 860, 728 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.80 (br s, 1 H), 8.21 (d, *J* = 7.8 Hz, 1 H), 8.17 (d, *J* = 7.8 Hz, 1 H), 7.58 (d, *J* = 7.6 Hz, 2 H), 7.46 (t, *J* = 7.8 Hz, 1 H), 7.38 (t, *J* = 7.6 Hz, 2 H), 7.27–7.31 (m, 1 H), 5.78 (br s, 1 H), 4.41 (qd, *J* = 7.1, 1.8 Hz, 2 H), 1.43 (t, *J* = 7.1 Hz, 3 H), 0.94 (s, 9 H), 0.12 (s, 3 H), 0.05 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 198.0, 165.8, 138.6, 134.6, 134.0, 133.6, 131.3, 130.6, 128.7, 128.3, 127.9, 125.6, 80.8, 61.2, 25.7, 18.2, 14.3, –5.0, –5.1.

MS (70 eV, EI): *m*/*z* (%) = 383 (5), 341 (55), 221 (100), 177 (5), 73 (78), 59 (8).

HRMS (EI): *m*/*z* [M – CN – Me – H]$^+$ calcd for $C_{22}H_{27}O_4Si$: 383.1679; found: 383.1646.

Ethyl 3-[(*tert*-Butyldimethylsilyloxy)dicyanomethyl]benzoate (6f)

According to GP3, compound **1a** (128 mg, 0.40 mmol) was dissolved in anhyd THF (1.6 mL). TMPMgCl·LiCl (**3**; 0.43 mL, 0.52 mmol, 1.21 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. A solution of tosyl cyanide (109 mg, 0.60 mmol) in anhyd THF (0.4 mL) was added at –78 °C and the mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **6f** as a colorless oil (69 mg, 50%).

IR (Diamond-ATR, neat): 2934, 2862, 2244, 1723, 1472, 1300, 1268, 1190, 1022, 837, 786, 688 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.38 (br s, 1 H), 8.20 (d, *J* = 7.8 Hz, 1 H), 7.87 (d, *J* = 7.8 Hz, 1 H), 7.62 (t, *J* = 7.8 Hz, 1 H), 4.43 (q, *J* = 7.1 Hz, 2 H), 1.42 (t, *J* = 7.1 Hz, 3 H), 1.00 (s, 9 H), 0.41 (s, 6 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 165.1, 135.3, 132.0, 132.0, 129.7, 129.3, 126.3, 114.8, 64.7, 61.6, 25.2, 18.2, 14.2, –4.4.

MS (70 eV, EI): *m*/*z* (%) = 344 (1), 299 (19), 287 (100), 242 (15), 232 (23), 177 (96), 149 (11), 75 (10).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{18}H_{24}N_2O_3Si$: 344.1556; found: 344.1551.

3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyanobut-3-enyl]benzonitrile (6g)

According to GP2, 3-[(*tert*-butyldimethylsilyloxy)(cyano)methyl]benzonitrile (**1b**; 136 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**, 1.57 mL, 0.55 mmol, 0.35 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.10 mL, 0.10 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. Allyl bromide (91 mg, 0.75 mmol) was added and the reaction mixture was stirred at –40 °C and the mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 96:4) yielding **6g** as a colorless oil (105 mg, 67%).

IR (Diamond-ATR, neat): 2931, 2859, 2233, 1644, 1584, 1472, 1362, 1257, 1108, 1096, 920, 837, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.83 (br s, 1 H), 7.78 (d, *J* = 7.8 Hz, 1 H), 7.67 (d, *J* = 7.8 Hz, 1 H), 7.54 (t, *J* = 7.8 Hz, 1 H), 5.63–5.72 (m, 1 H), 5.21 (d, *J* = 10.2 Hz, 1 H), 5.13 (d, *J* = 17.0 Hz, 1 H), 2.78 (dd, *J* = 13.7, 7.1 Hz, 1 H), 2.66 (dd, *J* = 13.7, 7.1 Hz, 1 H), 0.95 (s, 9 H), 0.26 (s, 3 H), 0.00 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 142.2, 132.4, 129.7, 129.6, 129.4, 128.9, 121.5, 119.5, 118.1, 112.9, 74.4, 50.1, 25.6, 18.3, –3.8, –3.8.

MS (70 eV, EI): *m*/*z* (%) = 297 (1), 271 (25), 255 (26), 228 (100), 130 (43), 99 (30), 84 (15), 75 (39), 73 (55), 57 (24), 41 (17).

HRMS (EI): *m*/*z* [M – Me]$^+$ calcd for $C_{18}H_{24}N_2OSi$: 297.1423; found: 297.1414.

3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyano-2-phenylethyl]benzonitrile (6h)

According to GP2, compound **1b** (82 mg, 0.30 mmol) was dissolved in anhyd THF (1.2 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 1.03 mL, 0.33 mmol, 0.32 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuBr[12] (4 mg, 0.03 mmol) and solid benzyl bromide (54 mg, 0.32 mmol) were added at –78 °C. The reaction mixture was stirred at –78 °C for 1 h and at 50 °C for 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 96:4) yielding **6h** as a white solid (80 mg, 74%); mp 97.7–98.7 °C.

IR (Diamond-ATR, neat): 2931, 2859, 2233, 1588, 1472, 1362, 1256, 1112, 1094, 908, 874, 839, 782, 693 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.73 (br s, 1 H), 7.64 (d, *J* = 7.8 Hz, 1 H), 7.66 (d, *J* = 7.8 Hz, 1 H), 7.47 (t, *J* = 7.8 Hz, 1 H), 7.23–7.27 (m, 3 H), 7.07 (d, *J* = 6.6 Hz, 2 H), 3.22 (d, *J* = 13.4 Hz, 1 H), 3.11 (d, *J* = 13.4 Hz, 1 H), 0.91 (s, 9 H), 0.01 (s, 3 H), –0.08 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 142.4, 133.0, 132.4, 130.9, 129.7, 129.3, 129.0, 128.1, 127.8, 119.5, 118.1, 112.7, 75.4, 52.1, 25.7, 18.3, –3.8, –4.2.

MS (70 eV, EI): *m*/*z* (%) = 362 (1), 305 (31), 271 (100), 265 (29), 255 (10), 130 (32), 90 (22), 84 (19), 75 (39), 73 (56), 41 (11).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{22}H_{26}N_2OSi$: 362.1814; found: 362.1817.

3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyano-3,3-dimethyl-2-oxobutyl]benzonitrile (6i)

According to GP2, compound **1b** (136 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 1.93 mL, 0.55 mmol, 0.28 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.10 mL, 0.10 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. Pivaloyl chloride (72 mg, 0.60 mmol) was added and the mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq. NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5) yielding **6i** as a light yellow solid (131 mg, 74%); mp 59.1–60.9 °C.

IR (Diamond-ATR, neat): 2933, 2861, 2233, 1719, 1472, 1363, 1258, 1151, 1095, 911, 840, 783, 688 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.85 (br s, 1 H), 7.80 (d, *J* = 7.8 Hz, 1 H), 7.71 (d, *J* = 7.8 Hz, 1 H), 7.56 (t, *J* = 7.8 Hz, 1 H), 1.28 (s, 9 H), 1.01 (s, 9 H), 0.33 (s, 3 H), 0.02 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 205.0, 139.0, 133.0, 130.7, 129.8, 129.7, 117.8, 117.8, 113.2, 79.8, 45.7, 27.3, 25.8, 18.5, –3.8, –3.8.

MS (70 eV, EI): *m*/*z* (%) = 299 (42), 272 (32), 188 (7), 130 (25), 85 (13), 75 (38), 73 (29), 58 (100), 41 (17).

HRMS (EI): *m*/*z* [M – C_4H_9]$^+$ calcd for $C_{16}H_{19}N_2O_2Si$: 299.1216; found: 299.1212.

3-{2-(*tert*-Butyldimethylsilyloxy)-2-cyano-2-[4-(dimethylamino)phenyl]ethyl}benzonitrile (6j)

According to GP2, 2-(*tert*-butyldimethylsilyloxy)-2-[4-(dimethylamino)phenyl]acetonitrile (**1c**; 145 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 1.93 mL, 0.55 mmol, 0.28 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuBr[12] (7 mg, 0.05 mmol) and solid 3-cyanobenzyl bromide (118 mg, 0.60 mmol) were added at –78 °C. The reaction mixture was stirred at –78 °C for 1 h and at 50 °C for 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **6j** as a white solid (168 mg, 83%); mp 117.6–119.1 °C.

IR (Diamond-ATR, neat): 2929, 2857, 2230, 1713, 1611, 1521, 1471, 1360, 1254, 1165, 1088, 946, 830, 779, 694 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.57 (d, *J* = 7.6 Hz, 1 H), 7.43–7.50 (m, 2 H), 7.40 (t, *J* = 7.6 Hz, 1 H), 7.32 (m, *J* = 8.9 Hz, 2 H), 6.69 (m, *J* = 8.9 Hz, 2 H), 3.28 (d, *J* = 13.5 Hz, 1 H), 3.13 (d, *J* = 13.5 Hz, 1 H), 3.00 (s, 6 H), 0.89 (s, 9 H), –0.02 (s, 3 H), –0.11 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 150.7, 136.2, 135.4, 134.7, 130.9, 128.7, 126.9, 126.3, 120.1, 118.7, 111.9, 111.7, 75.7, 51.7, 40.3, 29.7, 25.7, 18.3, –3.7, –4.4.

MS (70 eV, EI): *m*/*z* (%) = 405 (2), 378 (5), 289 (80), 274 (17), 262 (7), 247 (10), 148 (100), 73 (9), 57 (7).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{24}H_{31}N_3OSi$: 405.2236; found: 405.2229.

2-(*tert*-Butyldimethylsilyloxy)-3-cyclopropyl-2-[4-(dimethylamino)phenyl]-3-oxopropanenitrile (6k)

According to GP2, compound **1c** (145 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 1.57 mL, 0.55 mmol, 0.35 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.10 mL, 0.10 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. Cyclopropanecarbonyl chloride (78 mg, 0.75 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 98:2) yielding **6k** as a light yellow solid (152 mg, 85%); mp 73.3–75.1 °C.

IR (Diamond-ATR, neat): 2930, 2858, 2255, 1710, 1609, 1520, 1471, 1361, 1256, 1117, 1061, 1003, 939, 837, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.41 (d, *J* = 9.0 Hz, 2 H), 6.72 (d, *J* = 9.0 Hz, 2 H), 2.99 (s, 6 H), 2.19–2.27 (m, 1 H), 1.02–1.12 (m, 2 H), 1.01 (s, 9 H), 0.91–0.97 (m, 2 H), 0.26 (s, 3 H), 0.17 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 202.0, 151.0, 126.6, 122.7, 118.5, 112.1, 80.7, 40.2, 25.7, 18.4, 15.7, 13.1, 12.5, –4.0, –4.0.

MS (70 eV, EI): *m*/*z* (%) = 358 (1), 343 (2), 301 (16), 290 (15), 289 (66), 148 (100), 73 (9).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{20}H_{30}N_2O_2Si$: 358.2077; found: 358.2087.

Ethyl 2-(*tert*-Butyldimethylsilyloxy)-2-cyano-2-[4-(dimethylamino)phenyl]acetate (6l)

According to GP3, compound **1c** (145 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). TMPMgCl·LiCl (**3**; 0.65 mL, 0.59 mmol, 0.91 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. Ethyl cyanoformate (59 mg, 0.60 mmol) was added at –78 °C and the reaction mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/CH_2Cl_2, 9:1) yielding **6l** as a light yellow solid (161 mg, 89%); mp 47.1–48.8 °C.

IR (Diamond-ATR, neat): 2929, 2858, 2253, 1759, 1610, 1521, 1472, 1361, 1253, 1166, 1122, 1047, 945, 830, 782 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.49 (d, *J* = 8.4 Hz, 2 H), 6.71 (d, *J* = 8.4 Hz, 2 H), 4.14–4.30 (m, 2 H), 2.99 (s, 6 H), 1.26 (t, *J* = 7.1 Hz, 3 H), 0.98 (s, 9 H), 0.25 (s, 3 H), 0.15 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 167.5, 151.1, 126.5, 123.6, 118.4, 111.8, 75.0, 63.0, 40.2, 25.6, 18.4, 13.8, –4.2, –4.3.

MS (70 eV, EI): *m*/*z* (%) = 362 (1), 305 (22), 289 (33), 250 (11), 148 (100), 121 (34), 73 (14).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{19}H_{30}N_2O_3Si$: 362.2026; found: 362.2029.

2-(*tert*-Butyldimethylsilyloxy)-2-[4-(dimethylamino)phenyl]-2-(methylthio)acetonitrile (6m)

According to GP3, compound **1c** (145 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). TMPMgCl·LiCl (**3**; 0.65 mL, 0.59 mmol, 0.91 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. *S*-Methyl methanethiosulfonate (95 mg, 0.75 mmol) was added at –78 °C and the mixture was allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 98:2) yielding **6m** as a light yellow solid (124 mg, 74%); mp 52.8–53.7 °C.

IR (Diamond-ATR, neat): 2928, 2857, 2225, 1609, 1521, 1471, 1360, 1254, 1190, 1064, 946, 836, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, *J* = 8.8 Hz, 2 H), 6.70 (d, *J* = 8.8 Hz, 2 H), 3.00 (s, 6 H), 2.33 (s, 3 H), 0.96 (s, 9 H), 0.20 (s, 3 H), 0.02 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 151.0, 126.8, 125.1, 118.5, 111.4, 78.9, 40.2, 25.6, 18.3, 14.7, –3.9, –4.2.

MS (70 eV, EI): *m*/*z* (%) = 335 (0.2), 289 (40), 173 (7), 148 (100), 73 (17), 57 (8).

HRMS (EI): *m*/*z* [M – H]$^+$ calcd for $C_{17}H_{28}N_2OSSi$: 335.1613; found: 335.1601.

Ethyl 4-(*tert*-Butyldimethylsilyloxy)-4-cyano-2-methylene-4-(pyridin-3-yl)butanoate (6n)

According to GP2, 2-(*tert*-butyldimethylsilyloxy)-2-(pyridin-3-yl)acetonitrile (**1d**; 186 mg, 0.75 mmol) was dissolved in anhyd THF (3 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 2.4 mL, 0.83 mmol, 0.35 M in THF) was added dropwise at 0 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.15 mL, 0.15 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. Ethyl 2-(bromomethyl)acrylate[13] (173 mg, 0.90 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 8:2) yielding **6n** as a light yellow oil (165 mg, 60%).

IR (Diamond-ATR, neat): 2932, 2859, 2256, 1718, 1631, 1576, 1472, 1391, 1256, 1153, 1096, 915, 833, 781 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.82 (dd, *J* = 2.4, 0.8 Hz, 1 H), 8.62 (dd, *J* = 4.8, 1.6 Hz, 1 H), 7.84 (ddd, *J* = 8.0, 2.4, 1.6 Hz, 1 H), 7.33 (ddd, *J* = 8.0, 4.8, 0.8 Hz, 1 H), 6.40 (d, *J* = 1.2 Hz, 1 H), 5.77 (d, *J* = 1.2 Hz, 1 H), 4.00–4.11 (m, 2 H), 3.17 (dd, *J* = 13.6, 0.7 Hz, 1 H), 3.00 (dd, *J* = 13.6, 0.7 Hz, 1 H), 1.22 (t, *J* = 7.1 Hz, 3 H), 0.92 (s, 9 H), 0.22 (s, 3 H), –0.05 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 166.1, 150.2, 147.1, 135.8, 133.7, 133.2, 130.9, 123.0, 119.2, 74.0, 61.1, 46.2, 25.6, 18.2, 14.0, –3.7, –3.9.

MS (70 eV, EI): *m*/*z* (%) = 345 (3), 303 (100), 275 (96), 248 (52), 155 (14), 106 (67), 75 (42), 73 (48), 57 (9), 43 (27).

HRMS (EI): *m*/*z* [M – Me]$^+$ calcd for $C_{18}H_{25}N_2O_3Si$: 345.1634; found: 345.1627.

3-[2-(*tert*-Butyldimethylsilyloxy)-2-cyano-2-(pyridin-3-yl)ethyl]-benzonitrile (6o)

According to GP2, compound **1d** (128 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 1.60 mL, 0.55 mmol, 0.34 M in THF) was added dropwise at 0 °C and the reaction mixture was stirred for 2 h. CuBr[12] (7 mg, 0.05 mmol) and solid 3-cyanobenzyl bromide (118 mg, 0.60 mmol) were added at –78 °C. The reaction mixture was stirred at –78 °C for 1 h and at 50 °C for 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 8:2) yielding **6o** as a white solid (136 mg, 75%); mp 63.7–65.6 °C.

IR (Diamond-ATR, neat): 2931, 2859, 2231, 1712, 1576, 1420, 1362, 1256, 1097, 1025, 970, 830, 781, 694 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.81 (br s, 1 H), 8.68 (d, *J* = 3.5 Hz, 1 H), 7.79 (dt, *J* = 8.0, 1.8 Hz, 1 H), 7.59–7.64 (m, 1 H), 7.49 (s, 1 H), 7.40–7.47 (m, 2 H), 7.36 (dd, *J* = 8.0, 4.8 Hz, 1 H), 3.31 (d, *J* = 13.7 Hz, 1 H), 3.19 (d, *J* = 13.7 Hz, 1 H), 0.92 (s, 9 H), –0.01 (s, 3 H), –0.08 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 150.6, 146.9, 135.3, 134.9, 134.6, 132.9, 131.5, 129.0, 123.2, 118.9, 118.3, 112.4, 74.1, 51.5, 25.6, 18.3, –3.8, –4.3.

MS (70 eV, EI): *m*/*z* (%) = 348 (2), 306 (40), 279 (100), 247 (14), 106 (26), 73 (19).

HRMS (EI): *m*/*z* [M – Me]$^+$ calcd for $C_{20}H_{22}N_3OSi$: 348.1532; found: 348.1531.

2-(2-Bromopyridin-3-yl)-2-(*tert*-butyldimethylsilyloxy)-3-cyclopropyl-3-oxopropanenitrile (6p)

According to GP2, 2-(2-bromopyridin-3-yl)-2-(*tert*-butyldimethylsilyloxy)acetonitrile (**1e**; 326 mg, 1.00 mmol) was dissolved in anhyd THF (4 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 3.14 mL, 1.10 mmol, 0.35 M in THF) was added dropwise at 0 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.20 mL, 0.20 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. Cyclopropanecarbonyl chloride (125 mg, 1.20 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 20 mL), extracted with EtOAc (3 × 40 mL), and the organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 8:2) yielding **6p** as a white solid (357 mg, 91%); mp 107.9–109.7 °C.

IR (Diamond-ATR, neat): 2930, 2859, 1720, 1560, 1472, 1392, 1256, 1126, 1052, 976, 837, 807, 783 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.44 (dd, *J* = 4.7, 1.7 Hz, 1 H), 8.19 (dd, *J* = 7.8, 1.7 Hz, 1 H), 7.46 (dd, *J* = 7.8, 4.7 Hz, 1 H), 1.94–2.02 (m, 1 H), 1.18–1.28 (m, 2 H), 1.00–1.08 (m, 2 H), 0.98 (s, 9 H), 0.28 (s, 3 H), 0.22 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 198.9, 150.7, 140.4, 136.9, 133.9, 123.0, 116.0, 78.6, 25.5, 18.4, 16.8, 14.2, 13.4, –4.1, –4.1.

MS (70 eV, EI): *m*/*z* (%) = 394 (0.3), 339 (34), 337 (35), 328 (20), 326 (22), 186 (7), 184 (9), 69 (100), 43 (15), 41 (30).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{17}H_{23}BrN_2O_2Si$: 394.0712; found: 394.0703.

3-[1-(*tert*-Butyldimethylsilyloxy)-1-cyano-2-(2-iodophenyl)-2-oxoethyl]-2-chloroisonicotinonitrile (6q)

According to GP2, 3-[(*tert*-butyldimethylsilyloxy)(cyano)methyl]-2-chloroisonicotinonitrile (**1f**; 307 mg, 1.00 mmol) was dissolved in anhyd THF (4 mL). $TMP_2Zn{\cdot}2MgCl_2{\cdot}2LiCl$ (**2**; 3.14 mL, 1.10 mmol, 0.35 M in THF) was added dropwise at 0 °C and the reaction mixture was stirred for 1 h. CuCN·2LiCl[11] solution (0.20 mL, 0.20 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. 2-Iodobenzyol chloride (400 mg, 1.50 mmol) was added and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 20 mL), extracted with EtOAc (3 × 40 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 8:2) yielding **6q** as a light yellow solid (446 mg, 83%); mp 163.3–164.7 °C.

IR (Diamond-ATR, neat): 2932, 2860, 1712, 1572, 1470, 1363, 1264, 1114, 1018, 951, 828, 790 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.50 (d, *J* = 4.9 Hz, 1 H), 8.05–8.07 (m, 1 H), 7.73 (dd, *J* = 7.8, 1.3 Hz, 1 H), 7.70 (d, *J* = 4.9 Hz, 1 H), 7.17–7.22 (m, 1 H), 7.09 (td, *J* = 7.8, 1.3 Hz, 1 H), 0.96 (s, 9 H), 0.56 (s, 3 H), 0.50 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 187.2, 150.8, 150.3, 143.4, 134.0, 133.7, 132.7, 129.2, 128.7, 126.9, 121.3, 115.2, 115.1, 97.0, 79.2, 26.2, 19.1, –3.0, –3.2.

MS (70 eV, EI): *m*/*z* (%) = 522 (4), 480 (100), 232 (29), 203 (85), 104 (18), 73 (96), 57 (51).

HRMS (EI): *m*/*z* [M – Me]$^+$ calcd for $C_{20}H_{18}ClIN_3O_2Si$: 521.9901; found: 521.9888.

tert-Butyl 3-[1-(tert-Butyldimethylsilyloxy)-1-cyanobut-3-enyl]-1H-indole-1-carboxylate (6r)

According to GP2, *tert*-butyl 3-[(*tert*-butyldimethylsilyloxy)(cyano)methyl]-1*H*-indole-1-carboxylate (**1g**; 77 mg, 0.20 mmol) was dissolved in anhyd THF (0.8 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 0.65 mL, 0.22 mmol, 0.34 M in THF) was added dropwise at 0 °C and the reaction mixture was stirred for 2 h. CuCN·2LiCl[11] solution (0.04 mL, 0.04 mmol, 1.0 M in THF) was added at –40 °C and stirred for 5 min. Allyl bromide (36 mg, 0.30 mmol) was added at –40 °C and the reaction mixture was stirred at –40 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 5 mL), extracted with EtOAc (3 × 20 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 98:2) yielding **6r** as a colorless oil (61 mg, 72%).

IR (Diamond-ATR, neat): 2931, 2859, 1740, 1641, 1452, 1370, 1254, 1153, 1081, 922, 836, 746 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.21 (d, *J* = 8.2 Hz, 1 H), 7.74–7.79 (m, 2 H), 7.34–7.40 (m, 1 H), 7.26–7.30 (m, 1 H), 5.81 (ddt, *J* = 17.0, 10.2, 7.2 Hz, 1 H), 5.16–5.24 (m, 2 H), 3.06 (dd, *J* = 13.9, 7.2 Hz, 1 H), 2.91 (dd, *J* = 13.9, 7.2 Hz, 1 H), 1.70 (s, 9 H), 0.95 (s, 9 H), 0.28 (s, 3 H), –0.06 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 149.3, 136.3, 130.9, 126.3, 125.0, 123.7, 122.8, 120.7, 120.5, 120.5, 119.9, 115.5, 84.3, 71.0, 47.5, 28.1, 25.6, 18.4, –3.9.

MS (70 eV, EI): *m*/*z* (%) = 426 (2), 385 (35), 329 (100), 313 (10), 285 (43), 242 (11), 144 (59), 57 (40).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{24}H_{34}N_2O_3Si$: 426.2339; found: 426.2337.

tert-Butyl 3-{1-(tert-Butyldimethylsilyloxy)-1-cyano-2-hydroxy-2-[4-(methoxycarbonyl)phenyl]ethyl}-1H-indole-1-carboxylate (6s)

According to GP2, compound **1g** (193 mg, 0.50 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 1.62 mL, 0.55 mmol, 0.34 M in THF) was added dropwise at 0 °C and the reaction mixture was stirred for 2 h. Methy 4-formylbenzoate (123 mg, 0.75 mmol) was added at –78 °C and the mixture was stirred at –78 °C and allowed to warm up to 25 °C over 12 h. The resulting solution was quenched with sat. aq NH_4Cl (10 mL), extracted with EtOAc (3 × 30 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 96:4 to 9:1) yielding **6s** as a colorless oil (173 mg, 63%).

IR (Diamond-ATR, neat): 2954, 2858, 2256, 1724, 1658, 1539, 1450, 1371, 1274, 1144, 1054, 910, 837 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.76 (s, 1 H), 8.40 (dd, *J* = 6.6, 2.2 Hz, 1 H), 8.14 (d, *J* = 7.4 Hz, 1 H), 8.04 (d, *J* = 8.2 Hz, 2 H), 7.70 (d, *J* = 8.2 Hz, 2 H), 7.32–7.38 (m, 2 H), 5.60 (s, 1 H), 3.89 (s, 3 H), 1.71 (s, 9 H), 1.00 (s, 9 H), 0.15 (s, 3 H), 0.12 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 195.2, 166.7, 148.8, 144.6, 134.8, 134.7, 129.8, 129.5, 128.3, 125.4, 125.3, 124.4, 122.5, 115.0, 114.8, 85.1, 82.0, 52.0, 28.0, 25.8, 18.2, –4.9, –5.2.

MS (70 eV, EI): *m*/*z* (%) = 523 (0.4), 492 (1), 366 (23), 279 (29), 244 (17), 188 (51), 144 (100), 73 (55), 57 (16).

HRMS (EI): *m*/*z* [M – HCN]$^+$ calcd for $C_{29}H_{37}NO_6Si$: 523.2390; found: 523.2346.

2-(tert-Butyldimethylsilyloxy)-3-(2,6-dichlorophenyl)-2-(2,6-dimethoxypyrimidin-4-yl)propanenitrile (6t)

According to GP2, 2-(*tert*-butyldimethylsilyloxy)-2-(2,6-dimethoxypyrimidin-4-yl)acetonitrile (**1h**; 31 mg, 0.10 mmol) was dissolved in anhyd THF (2 mL). $TMP_2Zn·2MgCl_2·2LiCl$ (**2**; 0.31 mL, 0.11 mmol, 0.35 M in THF) was added dropwise at –20 °C and the reaction mixture was stirred for 2 h. CuBr[12] (1 mg, 0.01 mmol) and solid 2,6-dichlorobenzyl bromide (36 mg, 0.15 mmol) were added at –78 °C. The reaction mixture was stirred at –78 °C for 1 h and at 50 °C for 12 h. The resulting solution was quenched with sat. aq NH_4Cl/NH_3 solution (8:1, 5 mL), extracted with EtOAc (3 × 20 mL), and the combined organic phases were dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **6t** as a white solid (37 mg, 79%); mp 112.5–113.1 °C.

IR (Diamond-ATR, neat): 2955, 2859, 1633, 1571, 1461, 1355, 1206, 1126, 1094, 983, 780 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.30 (d, *J* = 8.0 Hz, 2 H), 7.15 (t, *J* = 8.0 Hz, 1 H), 6.62 (s, 1 H), 4.00 (s, 3 H), 3.92 (s, 3 H), 3.75–3.87 (m, 2 H), 0.91 (s, 9 H), 0.19 (s, 3 H), 0.14 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 172.5, 169.5, 165.0, 137.5, 130.7, 129.1, 128.3, 119.3, 97.9, 74.6, 55.1, 54.2, 43.5, 25.8, 18.3, –3.6, –3.9.

MS (70 eV, EI): *m*/*z* (%) = 467 (1), 412 (67), 410 (100), 307 (7), 251 (71), 167 (4).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{21}H_{27}Cl_2N_3O_3Si$: 467.1199; found: 467.1195.

Ethyl 3-[2-(4-Bromophenyl)acetyl]benzoate (7a)

According to GP4, ethyl 3-[2-(4-bromophenyl)-1-(*tert*-butyldimethylsilyloxy)-1-cyanoethyl]benzoate (**6d**; 207 mg, 0.43 mmol) was dissolved in THF (6.5 mL) and TBAF (0.47 mL, 0.47 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (20 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 40 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 95:5 to 9:1) yielding **7a** as a white solid (104 mg, 71%); mp 62.2–64.0 °C.

IR (Diamond-ATR, neat): 2983, 2905, 1716, 1688, 1488, 1300, 1276, 1194, 1012, 907, 802, 710, 682 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.65 (br s, 1 H), 8.21–8.28 (m, 1 H), 8.13–8.19 (m, 1 H), 7.55 (t, *J* = 7.7 Hz, 1 H), 7.45 (m, *J* = 8.4 Hz, 2 H), 7.15 (m, *J* = 8.4 Hz, 2 H), 4.42 (q, *J* = 7.2 Hz, 2 H), 4.28 (s, 2 H), 1.42 (t, *J* = 7.2 Hz, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 196.1, 165.6, 136.5, 134.0, 133.0, 132.4, 131.7, 131.2, 131.1, 129.5, 128.9, 121.0, 61.4, 44.8, 14.2.

MS (70 eV, EI): *m*/*z* (%) = 348 (1), 346 (1), 303 (4), 301 (4), 177 (100), 149 (13), 104 (7), 90 (5), 76 (8).

HRMS (EI): *m*/*z* [M$^+$] calcd for $C_{17}H_{15}BrO_3$: 346.0205; found: 346.0203.

3-(2-Phenylacetyl)benzonitrile (7b)

According to GP4, 3-[1-(*tert*-butyldimethylsilyloxy)-1-cyano-2-phenylethyl]benzonitrile (**6h**; 68 mg, 0.19 mmol) was dissolved in THF (2.9 mL) and TBAF (0.21 mL, 0.21 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (10 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **7b** as a white solid (34 mg, 81%); mp 76.6–78.2 °C.

IR (Diamond-ATR, neat): 3069, 2894, 2227, 1696, 1579, 1453, 1333, 1236, 1166, 1007, 998, 720 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.26 (s, 1 H), 8.21 (d, *J* = 7.8 Hz, 1 H), 7.81 (d, *J* = 7.8 Hz, 1 H), 7.58 (t, *J* = 7.8 Hz, 1 H), 7.31–7.37 (m, 2 H), 7.22–7.30 (m, 3 H), 4.28 (s, 2 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 195.5, 137.2, 136.0, 133.4, 132.5, 132.2, 129.7, 129.3, 128.9, 127.3, 117.8, 113.2, 45.6.

MS (70 eV, EI): *m*/*z* (%) = 221 (6), 130 (100), 102 (20), 91 (25), 75 (3), 65 (9), 51 (4).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{15}H_{11}NO$: 221.0841; found: 221.0841.

1-Cyclopropyl-2-[4-(dimethylamino)phenyl]ethane-1,2-dione (7c)

According to GP4, 2-(*tert*-butyldimethylsilyloxy)-3-cyclopropyl-2-[4-(dimethylamino)phenyl]-3-oxopropanenitrile (**6k**; 42 mg, 0.12 mmol) was dissolved in THF (1.8 mL) and TBAF (0.13 mL, 0.13 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (5 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 20 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **7c** as a yellow solid (21 mg, 81%); mp 121.4–123.2 °C.

IR (Diamond-ATR, neat): 2924, 2828, 1684, 1587, 1440, 1374, 1290, 1108, 906, 822, 728 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.89 (d, *J* = 9.2 Hz, 2 H), 6.67 (d, *J* = 9.2 Hz, 2 H), 3.09 (s, 6 H), 2.45–2.53 (m, 1 H), 1.26–1.32 (m, 2 H), 1.10–1.15 (m, 2 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 204.2, 190.7, 154.3, 132.6, 119.7, 110.9, 40.0, 18.8, 12.7.

MS (70 eV, EI): *m*/*z* (%) = 217 (7), 148 (100), 105 (5), 77 (4), 42 (4).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{13}H_{15}NO_2$: 217.1103; found: 217.1093.

Ethyl 2-[4-(Dimethylamino)phenyl]-2-oxoacetate (7d)

According to GP4, ethyl 2-(*tert*-butyldimethylsilyloxy)-2-cyano-2-[4-(dimethylamino)phenyl]acetate (**6l**; 107 mg, 0.30 mmol) was dissolved in THF (4.5 mL) and TBAF (0.33 mL, 0.33 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 1.5 h at –78 °C and diluted with H_2O (10 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 9:1) yielding **7d** as a white solid (57 mg, 86%); mp 87.8–89.6 °C.

IR (Diamond-ATR, neat): 2925, 1722, 1647, 1581, 1445, 1384, 1227, 1172, 1017, 823, 730 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.88 (d, *J* = 9.2 Hz, 2 H), 6.65 (d, *J* = 9.2 Hz, 2 H), 4.40 (q, *J* = 7.0 Hz, 2 H), 3.08 (s, 6 H), 1.40 (t, *J* = 7.0 Hz, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 184.1, 165.0, 154.4, 132.4, 120.1, 110.8, 61.7, 39.9, 14.1.

MS (70 eV, EI): *m*/*z* (%) = 221 (8), 149 (9), 148 (100), 77 (7), 42 (7).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{12}H_{15}NO_3$: 221.1052; found: 221.1046.

S-Methyl 4-(Dimethylamino)benzothioate (7e)

According to GP4, 2-(*tert*-butyldimethylsilyloxy)-2-[4-(dimethylamino)phenyl]-2-(methylthio)acetonitrile (**6m**; 118 mg, 0.35 mmol) was dissolved in THF (5.3 mL) and TBAF (0.39 mL, 0.39 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (10 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 98:2 to 96:4) yielding **7e** as a white solid (64 mg, 94%); mp 102.2–104.0 °C.

IR (Diamond-ATR, neat): 2927, 1638, 1595, 1445, 1369, 1242, 1167, 1065, 904, 820, 725 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.88 (d, *J* = 9.0 Hz, 2 H), 6.63 (d, *J* = 9.0 Hz, 2 H), 3.04 (s, 6 H), 2.43 (s, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 190.1, 153.5, 129.1, 124.7, 110.5, 39.9, 11.3.

MS (70 eV, EI): *m*/*z* (%) = 195 (1), 148 (12), 127 (11), 113 (14), 111 (11), 99 (13), 97 (16), 85 (42), 83 (20), 71 (51), 69 (29), 57 (100).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{10}H_{13}NOS$: 195.0718; found: 195.0697.

Ethyl 2-Methylene-4-oxo-4-(pyridin-3-yl)butanoate (7f)

According to GP4, ethyl 4-(*tert*-butyldimethylsilyloxy)-4-cyano-2-methylene-4-(pyridin-3-yl)butanoate (**6n**; 63 mg, 0.17 mmol) was dissolved in THF (2.6 mL) and TBAF (0.19 mL, 0.19 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (10 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 1:1) yielding **7f** as a colorless oil (32 mg, 83%).

IR (Diamond-ATR, neat): 2983, 1714, 1668, 1585, 1418, 1313, 1262, 1113, 1023, 973, 806, 703 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 9.20 (br s., 1 H), 8.79 (d, *J* = 4.4 Hz, 1 H), 8.25 (d, *J* = 7.7 Hz, 1 H), 7.43 (dd, *J* = 7.7, 4.4 Hz, 1 H), 6.43 (s, 1 H), 5.73 (s, 1 H), 4.20 (q, *J* = 7.0 Hz, 2 H), 3.99 (s, 2 H), 1.25 (t, *J* = 7.0 Hz, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 195.8, 166.1, 153.6, 149.8, 135.6, 134.2, 131.8, 128.9, 123.6, 61.1, 41.9, 14.1.

MS (70 eV, EI): *m*/*z* (%) = 219 (4), 174 (10), 106 (100), 78 (34), 57 (6), 45 (5), 43 (18).

HRMS (EI): *m*/*z* [M^+] calcd for $C_{12}H_{13}NO_3$: 219.0895; found: 219.0894.

3-[2-Oxo-2-(pyridin-3-yl)ethyl]benzonitrile (7g)

According to GP4, 3-[2-(*tert*-butyldimethylsilyloxy)-2-cyano-2-(pyridin-3-yl)ethyl]benzonitrile (**6o**; 90 mg, 0.25 mmol) was dissolved in THF (3.7 mL) and TBAF (0.28 mL, 0.28 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (10 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried (anhyd $MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 1:1) yielding **7g** as a light yellow solid (36 mg, 65%); mp 159.9–161.4 °C.

IR (Diamond-ATR, neat): 3055, 2925, 2231, 1690, 1585, 1419, 1338, 1266, 1194, 1005, 993, 802, 732 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 9.20–9.29 (m, 1 H), 8.83 (d, J = 3.3 Hz, 1 H), 8.28 (d, J = 8.0 Hz, 1 H), 7.55–7.64 (m, 2 H), 7.50–7.54 (m, 1 H), 7.44–7.50 (m, 2 H), 4.37 (s, 2 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 195.0, 154.0, 149.8, 135.7, 134.9, 134.2, 133.2, 131.5, 131.0, 129.5, 123.9, 118.5, 112.9, 44.8.

MS (70 eV, EI): m/z (%) = 222 (2), 116 (4), 106 (100), 89 (5), 78 (31), 51 (11).

HRMS (EI): m/z [M$^+$] calcd for $C_{14}H_{10}N_2O$: 222.0793; found: 222.0781.

1-(2-Bromopyridin-3-yl)-2-cyclopropylethane-1,2-dione (7h)

According to GP4, 2-(2-bromopyridin-3-yl)-2-(*tert*-butyldimethylsilyloxy)-3-cyclopropyl-3-oxopropanenitrile (**6p**; 115 mg, 0.29 mmol) was dissolved in THF (4.4 mL) and TBAF (0.32 mL, 0.32 mmol, 1.0 M) was added dropwise at –78 °C. The reaction mixture was stirred for 0.5 h at –78 °C and diluted with H_2O (10 mL). The resulting solution was allowed to warm to 25 °C and extracted with EtOAc (3 × 30 mL). The combined organic phases were dried ($MgSO_4$), filtered, and the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 8:2) yielding **7h** as a yellow oil (66 mg, 90%).

IR (Diamond-ATR, neat): 3009, 1689, 1572, 1443, 1388, 1287, 1119, 1055, 917, 805, 756 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.50 (dd, J = 4.7, 1.8 Hz, 1 H), 7.84 (dd, J = 7.6, 1.8 Hz, 1 H), 7.41 (dd, J = 7.6, 4.7 Hz, 1 H), 2.70–2.79 (m, 1 H), 1.31–1.40 (m, 2 H), 1.24–1.31 (m, 2 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 199.2, 191.4, 152.6, 139.4, 139.3, 134.1, 122.8, 17.5, 14.3.

MS (70 eV, EI): m/z (%) = 252 (1), 227 (19), 225 (15), 186 (35), 184 (39), 158 (24), 156 (26), 76 (36), 70 (15), 50 (22), 43 (22), 41 (100).

HRMS (EI): m/z [M$^+$] calcd for $C_{10}H_8BrNO_2$: 252.9738; found: 252.9719.

2-(2-Bromopyridin-3-yl)-3-cyclopropylpyrazine (8)

1-(2-Bromopyridin-3-yl)-2-cyclopropylethane-1,2-dione (**7h**; 60 mg, 0.24 mmol) was dissolved in EtOH (2 mL) and ethylenediamine (21 mg, 0.36 mmol) was added dropwise. The reaction mixture was stirred at 80 °C for 0.5 h and the solvents were evaporated in vacuo. The resulting oil was dissolved in $CHCl_3$ (2 mL), DDQ (109 mg, 0.48 mmol) was added, and the mixture was stirred at 70 °C for 5 h. Then, sat. aq $NaHCO_3$ (10 mL) was added and extracted with CH_2Cl_2 (3 × 30 mL). The combined organic phases were washed with sat. aq $NaHCO_3$ (2 × 10 mL) and dried (anhyd $MgSO_4$). After filtration, the solvents were evaporated in vacuo. The crude product was purified by flash column chromatography on silica gel (*i*-hexane/EtOAc, 1:1) yielding **8** as a light brown solid (21 mg, 32% over 2 steps); mp 56.6–58.2 °C.

IR (Diamond-ATR, neat): 3008, 2926, 2228, 1551, 1417, 1388, 1206, 1131, 1085, 902, 810, 733 cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.45–8.52 (m, 2 H), 8.40 (d, J = 2.1 Hz, 1 H), 7.75 (dd, J = 7.4, 2.1 Hz, 1 H), 7.44 (dd, J = 7.4, 4.9 Hz, 1 H), 1.68–1.78 (m, 1 H), 0.84–1.30 (m, 4 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 157.0, 150.9, 150.1, 143.9, 142.5, 140.1, 139.3, 136.7, 122.8, 14.1.

MS (70 eV, EI): m/z (%) = 276 (13), 274 (13), 196 (100), 169 (10), 142 (6), 76 (2).

HRMS (EI): m/z [M – H]$^+$ calcd for $C_{12}H_9BrN_3$: 273.9980; found: 273.9975.

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Supporting Information

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